

Chemistry of bleaching powder :-

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $\text{Ca}(\text{ClO})_2$.

It is a strong disinfectant and bleaching agent.

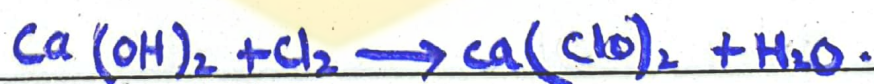
Composition :-

Bleaching powder is composed of :

- 1- calcium hypochlorite ($\text{Ca}(\text{ClO})_2$) : 30-35%.
- 2- calcium hydroxide ($\text{Ca}(\text{OH})_2$) : 30-40%.
- 3- calcium chloride (CaCl_2) : 10-20%.
- 4- water (H_2O) : 10-20%.

Preparation :-

Bleaching powder is prepared by reacting chlorine gas with slaked lime (calcium hydroxide).



Properties :-

- 1- Bleaching agent :- calcium hypochlorite is a strong oxidizing agent that breaks down chromophores, resulting in the removal

of color.

2- Disinfectant: Bleaching powder is effective against a wide range of microorganisms, including bacteria, viruses and fungi.

3- pH: Bleaching has a high pH (12-13) making it a strong alkaline substance.

Uses:-

1- Laundry: Bleaching clothes and removing stains.

2- Water treatment: Disinfecting and purifying water.

3- Swimming pools: Maintaining proper pH and disinfecting pool water.

4- Food industry: Sanitizing equipment and surfaces.

5- Medical: Disinfecting medical instruments and surfaces.

Shampoo :-

Shampoo is a hair product specifically formulated to cleanse the hair and scalp.

Ingredients :-

Surfactants: Cleansing agent that lower the surface tension of water allowing it to spread and penetrate more easily.

e.g. Sodium lauryl Sulfate (SLS) helps remove dirt and oil from hair.

Thickeners: Ingredients that increase the viscosity of a formulations. e.g.

Xanthan gum is often used in shampoos to create a desirable texture.

Preservatives: Chemical used to prevent microbial growth. e.g.

Parabens inhibit the growth of bacteria and mold.

Fragrances: Substance added to give a pleasant scent. e.g.

Essential oils like lavender or citrus provide fragrance.

Colourants: Dyes or pigments added.

e.g. Dand c yellow No. 5 is used to give Shampoo a yellow color-hue.

Function of Shampoo :-

- wash hair effectively.
- Remove dirt completely.
- Not have any side effects or irritation.
- To Prevent dandruff.

pH:-

- pH is a scale measurement of the acidity or basicity.
- Human hair has pH balance between 4.5 and 5.5.
- The pH of Shampoo usually ranges from 5 to 11.
- If the pH is too acidic, it will cause the cuticle damage and pores can may open.

Chemistry of Ceramics in

A ceramic is an inorganic non-metallic solid, generally based on an oxide, nitride, boride or carbide that is fired at high temperature.

Ceramics may be glazed prior to firing to produce a coating that reduces porosity and has smooth, often colored surface.

Raw materials in

Raw materials for ceramics include clay, kaolinite, aluminium oxide, silicon carbide, tungsten carbide and certain pure elements.

Properties of Ceramics in

- i- high hardness
- ii- High melting points
- iii- poor electrical and thermal conductivity
- iv- low ductility
- v- chemical resistance.

Types in

- 1- Earthenware: Made from clay and other natural materials, often used for pottery and tableware.

Stoneware: Made from combination of

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clay and other minerals, often used for cookware and decorative items.

Porcelain: Made from a combination of clay, feldspar and quartz, often used for fine china and decorative items.

Advanced ceramics: Made from a variety of materials including alumina, zirconia and silicon carbide, often used for industrial and chemical applications.

Applications in

Tableware and cookware: Ceramics are used to make plates, bowls, cups and cooking vessels.

Sanitary ware: Ceramics are used in toilets, sinks and bathroom fixtures.

Bricks & Tiles: Ceramics are used in construction of building walls, floors, roofs etc.

Medical: Ceramics are used in implants, such as hip and knee replacements, dental implants and surgical instruments.

Nuclear: Ceramics are used in nuclear

reactors and fuels due to high ^{thermal} radiation resistance.

Biotechnology: ceramics are used in bioreactors and tissue engineering scaffolds due to their high biocompatibility.

Textiles: Ceramics are used in textile manufacturing, such as in ceramic-coated fabrics and fibers.

Chemical Reactions:

During the ceramic manufacturing process various chemical reactions occur including:

1. Decomposition: Raw materials decompose into simpler compounds, releasing gases like water vapours and CO_2 .

2. Solid-state: Particles react with each other to form new compounds such as silicates and aluminates.

3. Vitrification: The formation of glassy phase, which can occur through the reaction of silica and alumina.

4. Crystallization: The formation of crystalline phases, such as quartz and feldspar.